

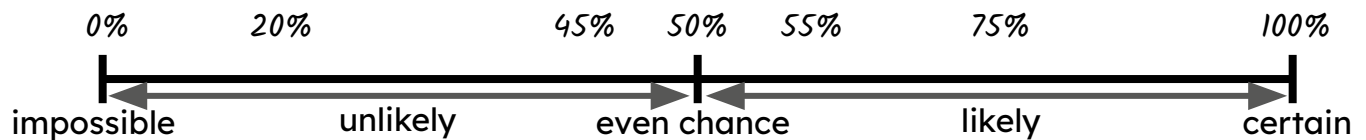


The probability scale

Task A: Expressing probabilities: percentages and decimals

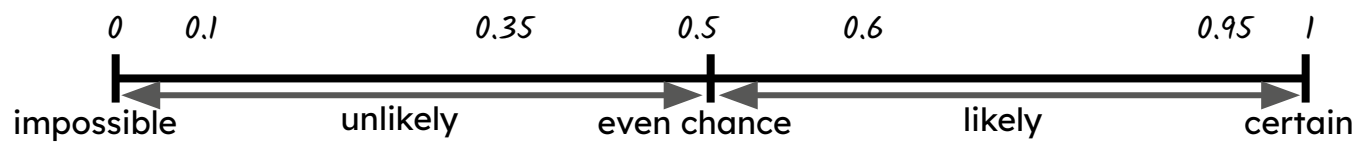
1) Write the **probabilities** below in the correct order on the **probability** scale.

0%, 100%, 20%, 75%, 50%, 45%, 55%



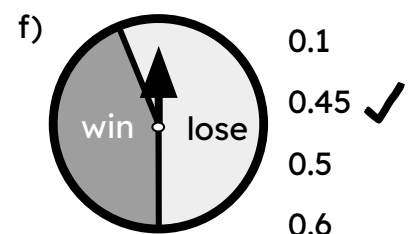
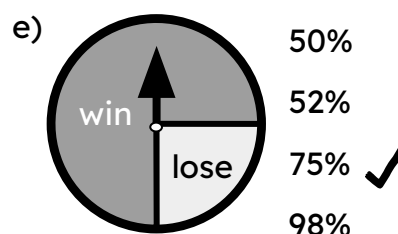
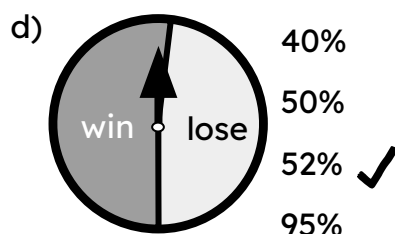
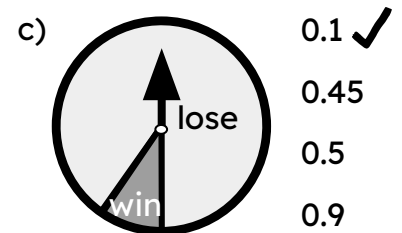
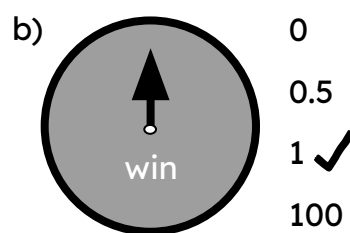
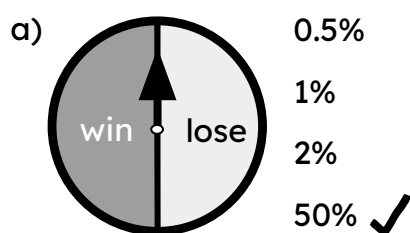
2) Write the **probabilities** below in the correct order on the **probability** scale.

1, 0, 0.5, 0.1, 0.35, 0.95, 0.6



3) In each question you are given a spinner and a choice of four **probabilities**.

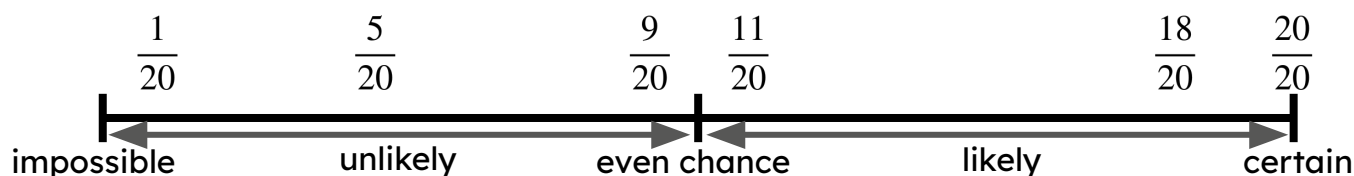
In each question, tick the **probability** that the spinner would land on 'win'.



Task B: Expressing probabilities as fractions

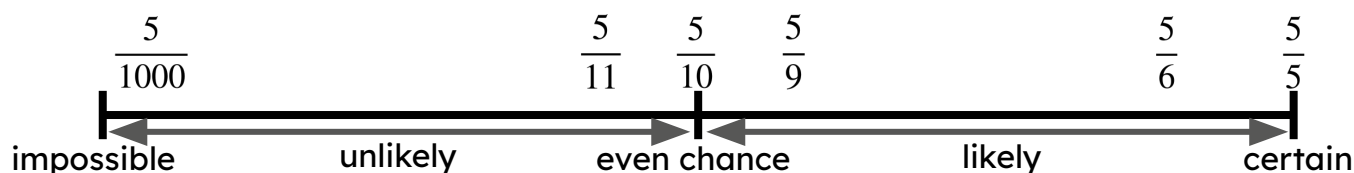
1) Write the **probabilities** below in the correct order on the **probability** scale.

$$\frac{20}{20} \quad \frac{1}{20} \quad \frac{5}{20} \quad \frac{18}{20} \quad \frac{11}{20} \quad \frac{9}{20}$$



2) Write the **probabilities** below in the correct order on the **probability** scale.

$$\frac{5}{6} \quad \frac{5}{5} \quad \frac{5}{1000} \quad \frac{5}{10} \quad \frac{5}{11} \quad \frac{5}{9}$$



3) Fill in the blanks in the table.

event	probability as a fraction	probability as a decimal	probability as a percentage	description of likelihood	expected frequency in 600 trials
example	$\frac{1}{2}$	0.5	50%	even chance	300
A	$\frac{1}{4}$	0.25	25%	unlikely	150
B	$\frac{3}{4}$	0.75	75%	likely	450
C	$\frac{2}{5}$	0.4	40%	unlikely	240
D	0	0	0%	impossible	0

4) A dropped cone could land in one of two ways: 'base down' or 'curve down'.

A large experiment concluded the following **probabilities**:



$$P(\text{base down}) = \frac{3}{10}$$



$$P(\text{curve down}) = \frac{7}{10}$$

a) The cone is about to be dropped again. Which outcome is most likely to happen?

curve down

b) In an experiment with 100 trials, approximately how many times could you expect the cone to land curve down?

$$\frac{7}{10} \text{ of } 100 = 70$$

c) Use the link to access a simulation of this experiment: [Cone simulation](#) 

The box that says 'count' shows the number of trials completed.

Press the 'run' button to begin the simulation and let it run for 100 trials.

Are the results the same as what you predicted in part (b)?

Example answers:

'The results were similar to my prediction but not the same.'

'There were 89 results for curve down, which is more than what I expected.'

'The number of times it landed curve down was much more than the number of times it landed base down, which was what I expected. But the exact number of times it landed on each side differed to what I expected.'

Answers may vary depending on results from the simulation.

